

Which Test, *When?*

A one-page decision guide for choosing the right hypothesis test

PARAMETRIC NON-PARAMETRIC
CATEGORICAL POST-HOC

STEP 1 · COMPARING NUMERIC OUTCOMES — PICK THE COLUMN FOR YOUR NUMBER OF GROUPS, THEN THE ROW FOR YOUR DATA

GROUPS → DATA ↓	One sample vs. a known / target value	Two groups independent samples	Three+ groups independent samples
Normal (parametric)	One-sample z-test if population σ is known One-sample t-test if σ unknown (usual case)	Two-sample z-test if both σ known / very large n Two-sample t-test if σ unknown (usual case)	One-way ANOVA compares all group means at once if significant → Tukey HSD
Not normal / ordinal	Wilcoxon signed-rank one sample or paired data; compares the median	Mann–Whitney U rank-based; two independent groups	Kruskal–Wallis H rank-based; three or more groups

STEP 2 · CATEGORICAL OUTCOMES (COUNTS & FREQUENCIES)

COUNTS	Chi-squared test of independence Use when you have two categorical variables and want to know if they are associated (related) or independent — e.g. is preference linked to region? Works on a contingency table of counts.
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QUICK REFERENCE — WHAT EACH TEST DOES & WHAT IT ASSUMES

TEST	USE IT WHEN...	KEY ASSUMPTIONS
One-sample z	Compare one sample mean to a known/target value, and the population SD σ is known.	σ known; data ~normal or $n \geq 30$; random sample.
Two-sample z	Compare means of two independent groups when both population SDs are known (or n is very large).	σ 's known; independent groups; large/normal.
One-sample t	Compare one sample mean to a known/target value when σ is unknown (estimated from the sample). The everyday case.	σ unknown; data approx. normal (matters most for small n).
Two-sample t	Compare means of two independent groups when σ is unknown.	Independent; approx. normal; equal variance (use Welch's if not).
Wilcoxon signed-rank	Non-parametric stand-in for the one-sample / paired t-test — compares a median when data isn't normal or is ordinal.	Paired or one sample; differences ~symmetric.
Mann–Whitney U	Non-parametric stand-in for the two-sample t-test — two independent groups, non-normal or ordinal data.	Independent; similar distribution shapes to compare medians.
One-way ANOVA	Compare the means of three or more independent groups in a single test (avoids inflating error from many t-tests).	Independent; approx. normal; equal variances.
Kruskal–Wallis H	Non-parametric stand-in for one-way ANOVA — three or more groups with non-normal or ordinal data.	Independent; similar shapes across groups.
Tukey HSD	Post-hoc test run after a significant ANOVA to find which specific pairs of groups differ, while controlling the family-wise error rate.	Only after sig. ANOVA; same ANOVA assumptions.
Chi-squared independence	Test whether two categorical variables are associated, using a table of observed counts.	Counts not %; expected cell counts ≥ 5 ; independent obs.

Z VS. T — THE ONE RULE THAT TRIPS PEOPLE UP

Reach for a **z-test** only when the population standard deviation σ is genuinely known (rare). In almost every real study σ is estimated from the data, so the **t-test** is the correct default. As sample size grows, t and z converge anyway.

EACH NON-PARAMETRIC TEST MIRRORS A PARAMETRIC ONE

One-sample / paired t → **Wilcoxon signed-rank**
Two-sample t → **Mann–Whitney U**
One-way ANOVA → **Kruskal–Wallis H**
Use the right-hand one when normality fails or data is ranked / ordinal.